Day 16 | Thursday

November 26, 2026

16

Hidden dimensions: square roots and formula checks

2:15 PM–3:30 PM • Math • 75 min

Week 5 • 1 scheduled block • 75 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

2:15 PM–3:30 PM • 75 min

Hidden dimensions: square roots and formula checks

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 5, Lesson 5

One math block today. Other scheduled program sessions remain separate.

Hidden dimensions: square roots and formula checks

Today’s goal: Find a missing dimension from a known area and check the answer in the original formula.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Exercise Book 8A Q23–24 (p.52); 8B Q6,13 (pp.53–54); 8C Q1–4 (p.55)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Recall and reconnect

Explain how to undo a square, and why dividing by two is not the same as taking a square root.

Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 8A Q23–24 (p.52); 8B Q6,13 (pp.53–54); 8C Q1–4 (p.55)

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Choose the operation that undoes the formula

1 For a square, $A = s^2$, so the side length is $s = \sqrt{A}$.

2 For a rectangle, $A = l \times w$; divide by the known side to find the other.

3 A square root undoes squaring; it is not the same as dividing by two.

4 A physical side length is reported as a positive length.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Validate a hidden dimension

1. Write the known area and its square unit.
2. Find the missing dimension using the appropriate inverse operation.
3. Multiply the found dimensions to reproduce the original area.
4. Report the dimension in a linear unit, not a square unit.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Divide for a rectangle; root for a square

RECTANGLE • $A = l \times w$
$A = 4{,}800\ cm^2$; $l = 80$ cm
$w = 4{,}800 \div 80 = 60$ cm
Check: $80 \times 60 = 4{,}800\ cm^2$.

SQUARE • $A = s^2$
$A = 3{,}136\ mm^2$
$s = \sqrt{3{,}136} = 56$ mm
Check: $56^2 = 3{,}136\ mm^2$.

Together • Work 8A Q23 and 8C Q1–2. Name the shape first.
Write the rearranged formula before calculator entry.
Demonstrate square-root entry for the learner's chosen calculator model.

Shape Canvas • Hide a square or rectangle dimension. Predict it on paper, then reveal it. Check the found dimensions by multiplication.

Textbook pp.131–134 • Exercise Book 8C Q1–2 p.55 • Units change from square units to length units

Textbook | page 132

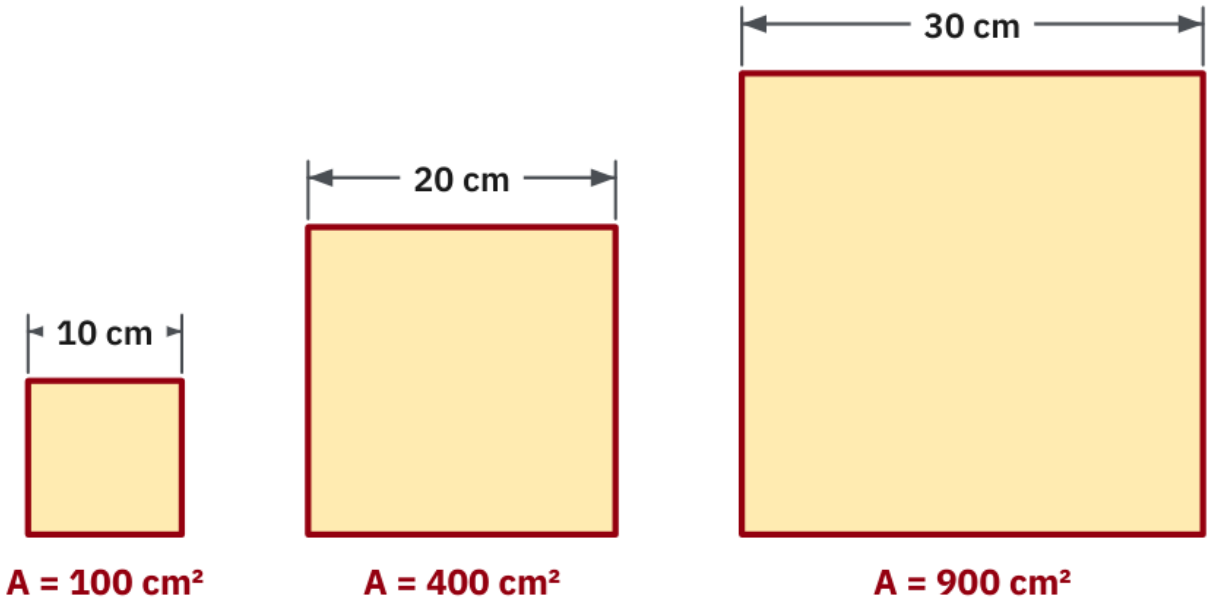


Figure 8.3 Change the side of a square and its area changes with it. Four times the area goes with twice the side. Nine times the area goes with three times the side.

Textbook p.132 • Figure 8.3 — Square side length and area

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Textbook | page 131

8.6 Formulas with squares and square roots

Squaring and taking the **square root** undo each other. If a letter has been squared, take the square root of both sides to get it back.

Example 8.11 Side of a square plate

A square plate must have an area of 62,500 mm². Transpose $A = s^2$ and find the side.

SOLUTION

1. s has been squared. Take the square root of both sides, so $s = \sqrt{A}$.
2. $s = \sqrt{62{,}500} = 250$

Textbook p.131 • Example 8.11 — Find a square side using a square root

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Spot the mistake. Explain the correction.

Someone divides a square’s area by two to find its side. Which inverse is needed?

Discuss the method before checking the answer.

Take the square root, because the side was multiplied by itself. Then square the result to check.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Work together

Work Exercise Book 8A Q23 and Exercise Book 8C Q1–2. Learners check by multiplying the found dimensions.

Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 8A Q23–24 (p.52); 8B Q6,13 (pp.53–54); 8C Q1–4 (p.55)

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Guided practice | 8A Q23

Make s the subject of $A = s^2$.

Work with a partner. Take turns explaining the next step.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Compare your method | 8A Q23

Make s the subject of $A = s^2$.

CHECK TOGETHER

$$s = \sqrt{A}$$

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Your scheduled practice

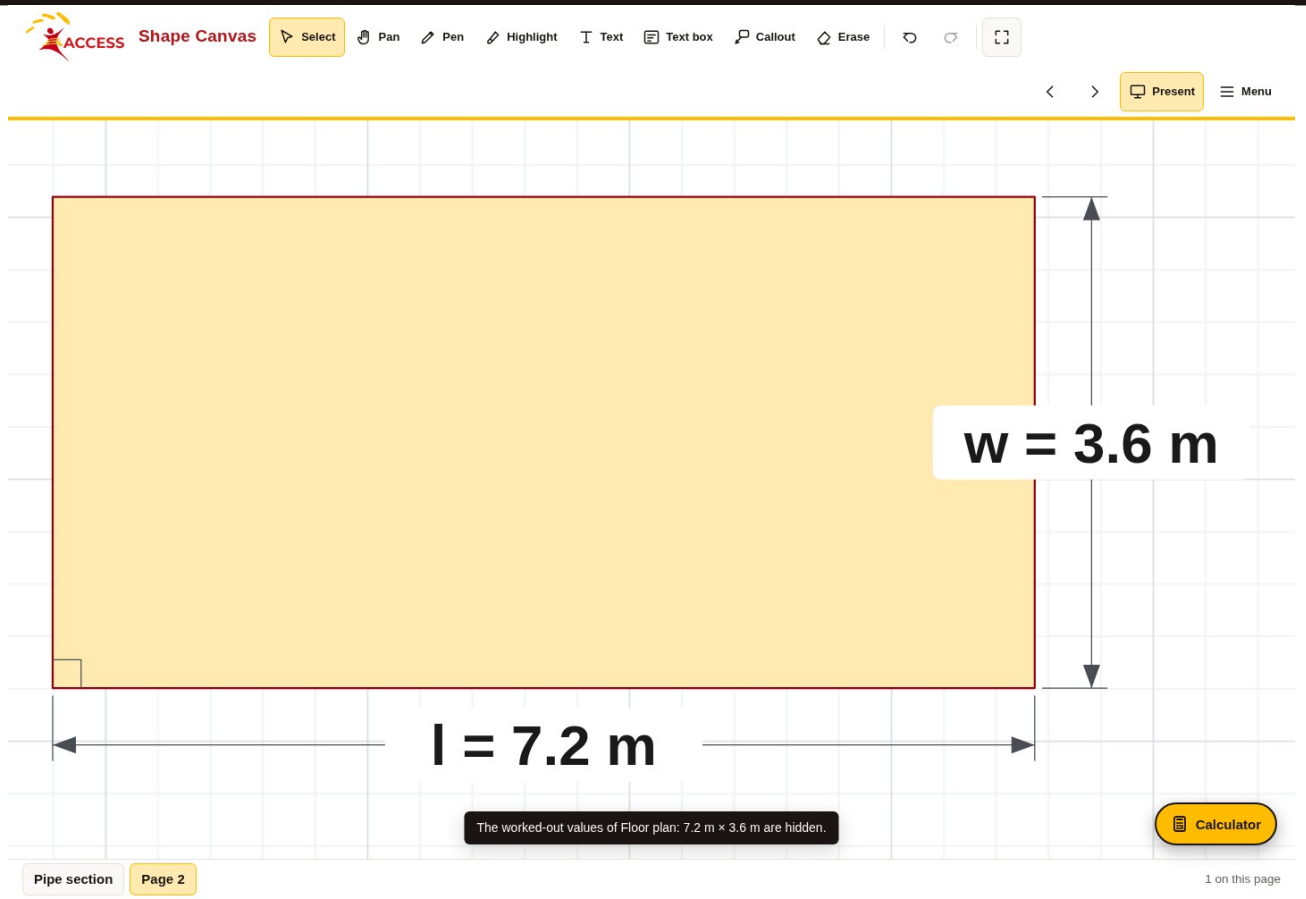
Complete Exercise Book 8B Q6,13; attempt Exercise Book 8C Q3–4 with formula support if ready. Make a hidden-dimension Shape Canvas prompt for a partner.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 8A Q23–24 (p.52); 8B Q6,13 (pp.53–54); 8C Q1–4 (p.55)

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Practise in Numeracy Hub and Shape Canvas

1. Create a square or rectangle in Shape Canvas and hide one dimension for a partner.

2. Calculate the hidden dimension on paper; use the selected calculator’s square-root key order.

3. Use Practice / Check all only after the attempt, then verify in the original area formula.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Practise from the Exercise Book | 8B Q6

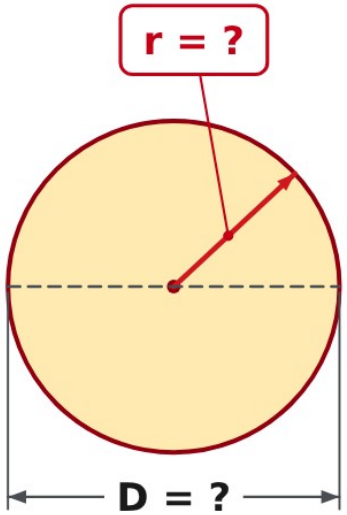
A square base plate has an area of 1,296 in². Find the side in inches and in feet.

Write the operation, working and units before checking.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Practise from the Exercise Book | 8C Q3

The circle has an area of 8,000 mm². Find r and D to two decimal places.



Write the operation, working and units before checking.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Check independently

Exercise Book 8C Q1–2 independently; substitute each answer back.

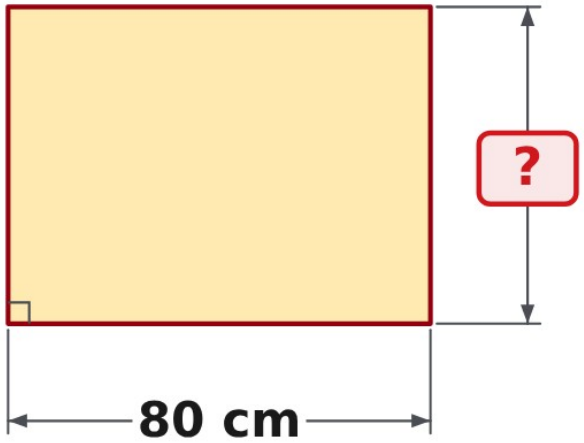
Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 8A Q23–24 (p.52); 8B Q6,13 (pp.53–54); 8C Q1–4 (p.55)

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Independent check | 8C Q1

The rectangle has an area of 4,800 cm². Find the width.

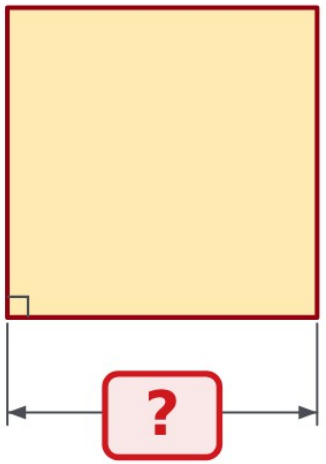


Answer independently. Keep worked solutions closed.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Independent check | 8C Q2

The square has an area of 3,136 mm². Find the side.



Answer independently. Keep worked solutions closed.

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 8A Q23–24 (p.52); 8B Q6,13 (pp.53–54); 8C Q1–4 (p.55)

Textbook sections 8.6 pp.131–133; sections 8.7 p.134; calculator pp.31–33